

tf.compat.v1.reduce_sum Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/reduce_sum

Computes the sum of elements across dimensions of a tensor. (deprecated arguments)

Function Signatures

```
tf.compat.v1.reduce_sum( input_tensor, axis=None,
keepdims=None, name=None, reduction_indices=None,
keep_dims=None )
(keep_dims)
```

Code Examples

```
tf.compat.v1.reduce_sum(
input_tensor,
axis=None,
keepdims=None,
name=None,
reduction_indices=None,
keep_dims=None
)
```

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input_tensor,  
axis=None,  
keepdims=None,  
name=None,  
reduction_indices=None,  
keep_dims=None  
)
```

```
>>> # x has a shape of (2, 3) (two rows and three columns):  
>>> x = tf.constant([[1, 1, 1], [1, 1, 1]])  
>>> x.numpy()  
array([[1, 1, 1],  
       [1, 1, 1]], dtype=int32)  
>>> # sum all the elements  
>>> # 1 + 1 + 1 + 1 + 1 + 1 = 6  
>>> tf.reduce_sum(x).numpy()  
6  
>>> # reduce along the first dimension  
>>> # the result is [1, 1, 1] + [1, 1, 1] = [2, 2, 2]  
>>> tf.reduce_sum(x, 0).numpy()  
array([2, 2, 2], dtype=int32)  
>>> # reduce along the second dimension  
>>> # the result is [1, 1] + [1, 1] + [1, 1] = [3, 3]  
>>> tf.reduce_sum(x, 1).numpy()  
array([3, 3], dtype=int32)  
>>> # keep the original dimensions  
>>> tf.reduce_sum(x, 1, keepdims=True).numpy()  
array([[3],  
       [3]], dtype=int32)  
>>> # reduce along both dimensions  
>>> # the result is 1 + 1 + 1 + 1 + 1 + 1 = 6  
>>> # or, equivalently, reduce along rows, then reduce the  
resultant array  
>>> # [1, 1, 1] + [1, 1, 1] = [2, 2, 2]  
>>> # 2 + 2 + 2 = 6  
>>> tf.reduce_sum(x, [0, 1]).numpy()  
6
```

```

>>> # x has a shape of (2, 3) (two rows and three columns):
>>> x = tf.constant([[1, 1, 1], [1, 1, 1]])
>>> x.numpy()
array([[1, 1, 1],
       [1, 1, 1]], dtype=int32)
>>> # sum all the elements
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>>> tf.reduce_sum(x).numpy()
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>>> # reduce along the first dimension
>>> # the result is [1, 1, 1] + [1, 1, 1] = [2, 2, 2]
>>> tf.reduce_sum(x, 0).numpy()
array([2, 2, 2], dtype=int32)
>>> # reduce along the second dimension
>>> # the result is [1, 1] + [1, 1] + [1, 1] = [3, 3]
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array([3, 3], dtype=int32)
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>>> tf.reduce_sum(x, 1, keepdims=True).numpy()
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       [3]], dtype=int32)
>>> # reduce along both dimensions
>>> # the result is 1 + 1 + 1 + 1 + 1 + 1 = 6
>>> # or, equivalently, reduce along rows, then reduce the
>>> # resultant array
>>> # [1, 1, 1] + [1, 1, 1] = [2, 2, 2]
>>> # 2 + 2 + 2 = 6
>>> tf.reduce_sum(x, [0, 1]).numpy()
6

```

Documentation

Computes the sum of elements across dimensions of a tensor. (deprecated arguments)

Used in the notebooks

- Fitting Dirichlet Process Mixture Model Using Preconditioned Stochastic Gradient Langevin Dynamics
 - Universal Sentence Encoder-Lite demo
 - Linear Mixed-Effect Regression in {TF Probability, R, Stan}

This is the reduction operation for the elementwise `tf.math.add` op.

Reduces `input_tensor` along the dimensions given in `axis`.

Unless `keepdims` is true, the rank of the tensor is reduced by 1 for each of the entries in `axis`, which must be unique. If `keepdims` is true, the reduced dimensions are retained with length 1.

If `axis` is `None`, all dimensions are reduced, and a tensor with a single element is returned.

For example

Returns

numpy compatibility

Equivalent to `np.sum` apart the fact that numpy upcast `uint8` and `int32` to `int64` while tensorflow returns the same dtype as the input.

